

# Short-Range FM RADIO STATION

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**T**his article shows you how to build a small radio station at home and share music with others. The station can also be used for making announcements in colleges, industries, hospitals, schools and other places using a condenser mic amplifier circuit.

In this project an FM transmitter V2.0 module is used, as shown in Fig. 1. It is available on [www.ebay.com](http://www.ebay.com) and [www.amazon.in](http://www.amazon.in). The module can modulate your voice or music on FM radio. If you have a radio receiver, you can receive and listen to it.

This module communicates with Arduino Uno R3 or another micro-controller through I2C interface. All commands are sent through I2C interface. Here, Arduino Uno R3 along with Arduino library file is used for giving commands to control and vary different frequencies within 88MHz - 108MHz FM band.

## Circuit and working

The FM transmitter V2.0 module has the following four pins:

1. Vcc (5V)
2. GND
3. SDA (serial data I2C pin)
4. SCK (serial clock I2C pin)

A standard 3.5mm male-to-male audio cable (Fig. 2) is used for connecting with the MP3 player, mobile or computer.

Block diagram of the short-range FM station is shown in Fig. 3 and its circuit diagram in Fig. 4. It is built around Arduino Uno R3 (BOARD1), FM transmitter V2.0 module (TX1) and a few other components.

A simple fmtx library is used. It is an open source library, available at <https://docs.google.com/file/>

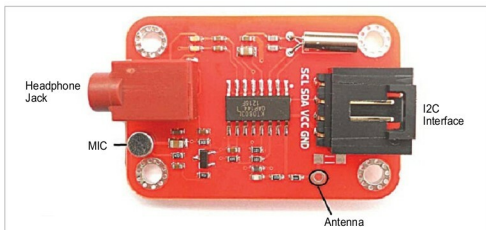


Fig. 1: FM transmitter V2.0 module



Fig. 2: 3.5mm male-to-male audio cable

## PARTS LIST

|                        |                                        |
|------------------------|----------------------------------------|
| <b>Semiconductors:</b> |                                        |
| BOARD1                 | - Arduino Uno R3                       |
| TX1                    | - FM transmitter V2.0 module           |
| <b>Miscellaneous:</b>  |                                        |
|                        | - 3.5mm male-to-male audio cable       |
|                        | - 9V battery or 9V adaptor             |
|                        | - 75cm single-strand copper antenna    |
|                        | - FM radio receiver                    |
|                        | - MP3 player or any other audio source |

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## Setting up input in FM band

**Step 1.** Download fmtx.cpp and fmtx.h libraries, and save these on your PC under, say, ftx folder. Copy this folder to Arduino\libraries folder.

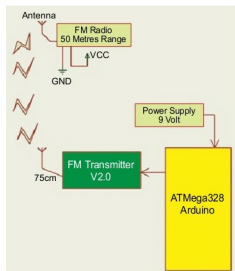


Fig. 3: Block diagram of FM station

**Step 2.** Open fmtx\_demo.ino code from Arduino IDE. Compile it and upload the program/sketch to Arduino board.

**Step 3.** Connect Arduino board to an audio source like mobile phone or MP3 player. You can also use a mic to transmit your voice.

**Step 4.** Turn on your FM receiver and set it to 90.1MHz frequency.

**Step 5.** Open Serial Monitor window from Arduino IDE to change the frequency, as shown in Fig. 5.

**Step 6.** You can change or set a